

BOISE, ID, USA

WMO: 726810

Lat: 43.567N

Lon: 116.241W

Elev: 2814

StdP: 13.26

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.4	16.4	3.6	7.3	14.7	8.3	9.3	22.6	24.8	43.3	22.2	41.4	4.5	130	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.4	98.7	63.7	95.9	62.9	93.1	61.9	66.1	92.0	64.7	90.4	63.4	88.6	9.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	78.3	71.2	55.1	71.6	71.8	52.9	66.2	70.9	32.1	91.4	31.1	90.2	30.0	88.4	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	19.1	17.0	6.6	104.3	7.0	2.7	1.6	106.2	-2.4	107.8	-6.4	109.4	-11.4	111.3
			5.7	68.6	6.5	1.8	1.0	69.9	-2.8	71.0	-6.4	72.0	-11.2	73.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	32.4	37.4	45.1	50.7	59.4	68.0	77.9	76.0	66.3	53.0	40.9	32.1
	DBStd	17.30	7.90	6.91	6.67	7.12	8.01	7.81	6.10	6.17	7.85	7.92	8.20	8.36
	HDD50	2071	547	354	181	76	13	0	0	0	2	56	288	554
	HDD65	5311	1012	772	617	432	210	57	3	5	79	381	724	1019
	CDD50	3297	0	2	29	96	306	539	866	805	491	148	14	1
	CDD65	1062	0	0	0	3	38	146	404	345	118	8	0	0
	CDH74	14022	0	0	2	68	563	1991	5487	4370	1430	111	0	0
	CDH80	7110	0	0	0	13	179	903	3110	2307	573	25	0	0
	WSAvg	7.4	6.4	7.7	8.6	8.7	8.0	7.9	7.2	7.0	6.9	6.9	6.8	7.0
	PrecAvg	11.6	1.4	1.0	1.3	1.2	1.2	0.7	0.3	0.3	0.6	0.8	1.3	1.4
(15) Precipitation	PrecMax	18.8	3.9	3.7	3.5	2.8	4.4	2.1	1.6	2.4	2.9	2.6	3.3	4.2
	PrecMin	6.7	0.2	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1
	PrecStd	2.6	0.8	0.7	0.8	0.6	1.0	0.6	0.4	0.4	0.6	0.6	0.7	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	59.7	71.1	81.5	90.5	98.7	103.6	101.0	94.8	83.4	65.6	56.2
		MCWB	44.4	47.0	50.4	55.9	60.0	63.9	65.2	63.6	62.3	56.4	49.8	46.0
	2%	DB	48.5	54.9	65.9	74.5	85.0	93.7	99.7	97.4	90.3	76.7	60.0	51.1
		MCWB	41.9	44.8	48.5	53.3	58.7	61.5	63.9	63.2	60.7	54.6	47.9	44.0
	5%	DB	45.4	51.3	61.4	69.4	80.4	89.3	97.2	94.7	86.6	71.4	56.3	47.1
		MCWB	40.0	42.4	46.9	51.0	56.6	60.4	63.4	62.3	59.3	52.7	46.0	41.4
	10%	DB	42.7	48.2	57.4	64.5	75.8	85.3	94.2	91.8	82.7	66.6	52.6	43.4
		MCWB	38.2	40.9	45.0	49.0	55.4	59.1	62.5	61.4	57.8	50.5	44.1	38.8
	0.4%	WB	46.2	48.5	52.0	57.3	62.5	66.2	68.1	66.9	66.3	59.0	52.7	47.5
		MCDB	50.5	57.1	65.3	77.0	84.2	93.0	94.6	93.3	83.7	77.0	61.4	53.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.6	45.5	50.1	54.5	60.1	63.7	66.4	65.3	63.3	56.2	49.7	44.6
		MCDB	47.2	53.5	62.4	71.1	79.8	88.4	93.2	91.9	82.5	72.6	57.5	50.7
	5%	WB	40.6	43.5	48.1	52.2	58.3	61.8	65.2	63.9	61.1	53.7	47.2	42.0
		MCDB	44.8	50.3	59.6	66.4	76.5	84.8	91.9	90.1	82.3	68.9	54.7	46.4
	10%	WB	38.5	41.5	46.2	50.1	56.5	60.1	63.9	62.5	59.1	51.4	44.9	39.4
		MCDB	42.3	47.2	56.3	63.0	73.3	81.7	90.0	88.1	80.5	64.6	51.8	43.1
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	15.6	19.3	21.4	23.4	26.0	29.4	28.2	25.8	21.8	16.0	12.0
		MCDBR	14.5	18.9	25.1	28.7	30.1	31.8	31.8	30.9	29.8	27.0	20.0	14.4
		MCWBR	9.7	11.1	13.0	13.2	11.6	11.1	11.1	11.4	12.2	12.1	11.1	9.4
	5% WB	MCDBR	13.6	17.5	22.0	25.6	26.8	29.0	29.2	28.2	27.0	24.1	17.9	13.2
		MCWBR	9.5	10.8	12.0	12.3	10.9	10.5	10.4	10.8	11.1	11.5	10.5	9.1
(40) Clear-Sky Solar Irradiance	taub		0.262	0.272	0.285	0.310	0.330	0.327	0.339	0.341	0.317	0.290	0.274	0.264
	taud		2.559	2.585	2.570	2.489	2.453	2.490	2.463	2.437	2.507	2.580	2.568	2.556
	Ebn at Noon		284	300	306	304	298	298	293	290	290	288	276	272
	Edh at Noon		21	24	28	32	35	34	34	34	29	24	20	19
(44) All-Sky Solar Radiation	RadAvg		512	830	1240	1728	2101	2396	2456	2117	1645	1078	630	420
	RadStd		69	108	126	137	151	129	91	93	91	80	51	60

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	-2.20	N/A	N/A	N/A	N/A	N/A	+164	+113
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page